

SGA 2: Local Cohomology and Lefschetz Theorems

Exposé VIII: The Finiteness Theorem

Bounded source-aligned working unit — 19 July 2026

Authority. Corrected French lines 2597–2609; original printed pp. 87–88; physical source-PDF p. 77; recomposed running p. 69. This bounded unit is translated and self-audited against the French TeX and direct PDF. Independent source review and cumulative integration remain pending. Line 2610 is blank; the next proof computation begins at line 2611 and is excluded.

Let $A^\bullet$ then be the complex defined by $A^p = 0$ if $p \neq 0$ and $A^0 = A$. Let

$$a: A^\bullet \longrightarrow CA^\bullet$$

be an injective resolution of $A^\bullet$. Consider the double complex

$$Q^{p,q} = \operatorname{Hom}(\operatorname{Hom}(L^{-q}, A), CA^p). \quad (1.2)$$

The first spectral sequence of the bicomplex $\underline{E}Q^{\bullet\bullet}$ will give the conclusion of Lemma 1.2.

Set

$$L'^\bullet = \operatorname{Hom}^\bullet(L^\bullet, A^\bullet), \quad (1.3)$$

and

$$P^\bullet = \operatorname{Hom}^\bullet(L'^\bullet, CA^\bullet). \quad (1.4)$$